OOPS Banner App

OOPS Banner App starts with the simple use case of displaying OOPS in the console to extending the use case to display in a Banner Format using space and * and goes on to refactor the code to make it more modular using functions then defining a class to store the character pattern and finally using a map to store the character pattern and use function to display the OOPS in a Banner Format

UC2: Render OOPS as Banner using Print Statements

Goal: Extend UC1 by displaying "OOPS" in a large banner made from spaces and `*` characters using individual print statements for each line.
.
Actor: User

Flow: User runs the program; program prints multiple lines of `*` and spaces to form each letter (O, O, P, S); program exits.

Key Concepts used in UC2:

1. **ASCII Art —** Creating visual representations using characters

2. **String Manipulation —** Constructing strings to form patterns using concatenation

3. **Operator Overloading —** Using the + operator for string concatenation

4. **Multiple Print Statements –** Sequential System.out.println() calls to build multi-line banner output

Key Requirements for UC2:

1. Use asterisks (*) and spaces( ) to form each letter

2. Print each line of the banner individually

3. Each Character should be represented in a 7-line format and a width of 9 characters, aligned properly

4. Use string concatenation + to add 2 strings for pattern creation

5. Use the hint below to guide the construction of each line

6. Ensure the output visually represents the letters O, O, P, S

Please refer to the hint below to write your code

```
// Hint: Use the Example below as a guide to construct each line of the banner
// Example for OS which stands for Operating System:
System.out.println("   ***   " + " " + "   ***** ");
System.out.println(" **   ** " + " " + " **      ");
System.out.println("**     **" + " " + "**       ");
System.out.println("**     **" + " " + "  *****  ");
System.out.println("**     **" + " " + "      ** ");
System.out.println(" **   ** " + " " + " **   ** ");
System.out.println("   ***   " + " " + "  *****  ");
```

Please refer to the code snapshot below to write your code

```java
/**
 * OOPSBannerApp UC2 - Banner Display Application
 *
 * This class demonstrates the creation of a visual banner displaying the word "OOPS"
 * using ASCII art composed of asterisks (*) and spaces. The banner is printed to the
 * console using individual print statements for each line, creating a decorative
 * text effect.
 *
 * <p>The banner is generated by printing predefined string patterns line by line,
 * where each line contributes to forming the letters O, O, P, and S in a stylized format.
 * This extends a basic user story by adding visual presentation to a simple message output.
 * </p>
 *
 * @author Developer
 * @version 2
 */

// Extend the User Story 1 to display the message "OOPS" in a banner format
// using characters formed by spaces and asterisks (*) to create a visual
// effect for each letter in the message. Use print statements for each line
// instead of a loop.

public class OOPSBannerApp {

    // Main method to run the banner display
    // Print each line of the OOPS banner to create the visual effect for the
    // message "OOPS". Construct each line using string concatenation.
    // Each line corresponds to a row in the banner for the letters O, O, P, S.
    // Using individual print statements for each line.
    Run | Debug
    public static void main(String[] args) {…

}
```

Please refer to the output after compiling and running the Program

```
!javac OOPSBannerApp.java
!java OOPSBannerApp
```

```
...     ***       ***    ******       *****
      **   **   **   **  **    **   **
     **     ** **     ** **     ** **
     **     ** **     ** **    **   **
     **     ** **     ** ******       ***
     **     ** **     ** **              **
     **     ** **     ** **               **
      **   **   **   **  **              **
        ***       ***    **         *****
```

[OOPS Banner App - UC2 - GenAI Video - Print Banner]

OOPS Banner App - UC2 - GenAI Audio - Print Banner